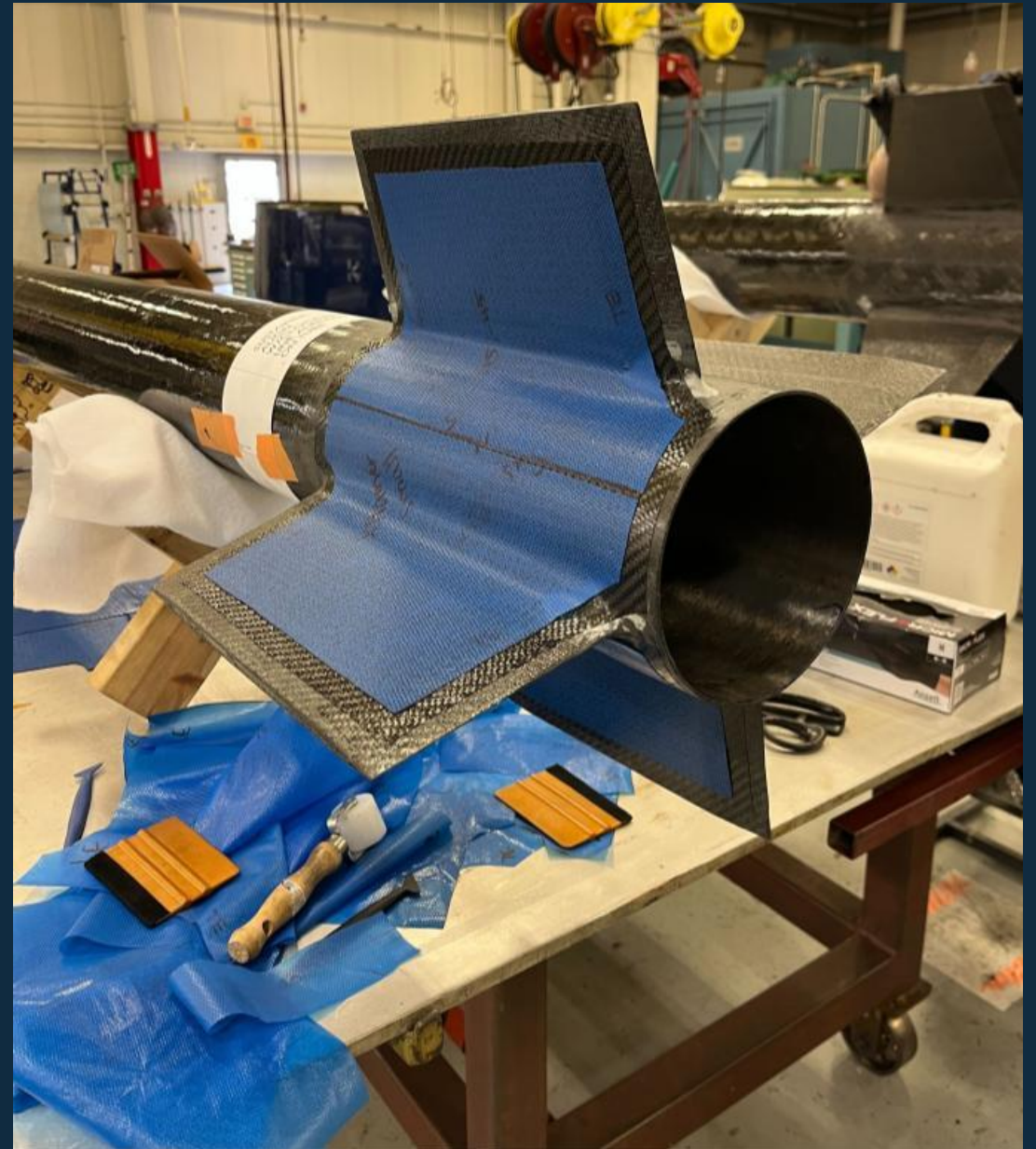

NATHANIEL MURCIN

Aerospace Engineering

Georgia Institute of Technology

Engineering portfolio



GTXR fin layup

Education

Georgia Institute of Technology

B.S. Aerospace Engineering

Expected May 2027

B.S./M.S. Aerospace Engineering Program

Minor in Astrophysics

GPA 4.00

Capability areas

Fluid systems and aerospace hardware

Structural analysis and experimental testing

Precision tooling and manufacturing

Experience

Blue Origin May 2026 - Aug. 2026

Pursuit Aerospace Aug. 2025 - Dec. 2025

Georgia Tech Experimental Rocketry Aug. 2023 - Present

Ben T. Zinn Combustion Laboratory Jan. 2026 - Present

Professional aerospace experience

03

BLUE ORIGIN

Fluids & Manufacturing Engineering Intern

Cape Canaveral, FL / May 2026 - Aug. 2026

FLUIDS ENGINEERING

Designed a cryogenic tank-farm P&ID for a large-scale ground-test facility. Reduced component count about 60% versus heritage designs by downsizing, eliminating branches, and challenging requirements.

Sized, selected, and validated cryogenic piping, valves, and relief devices. Developed the BOM and coordinated with suppliers through procurement.

LUNAR MANUFACTURING

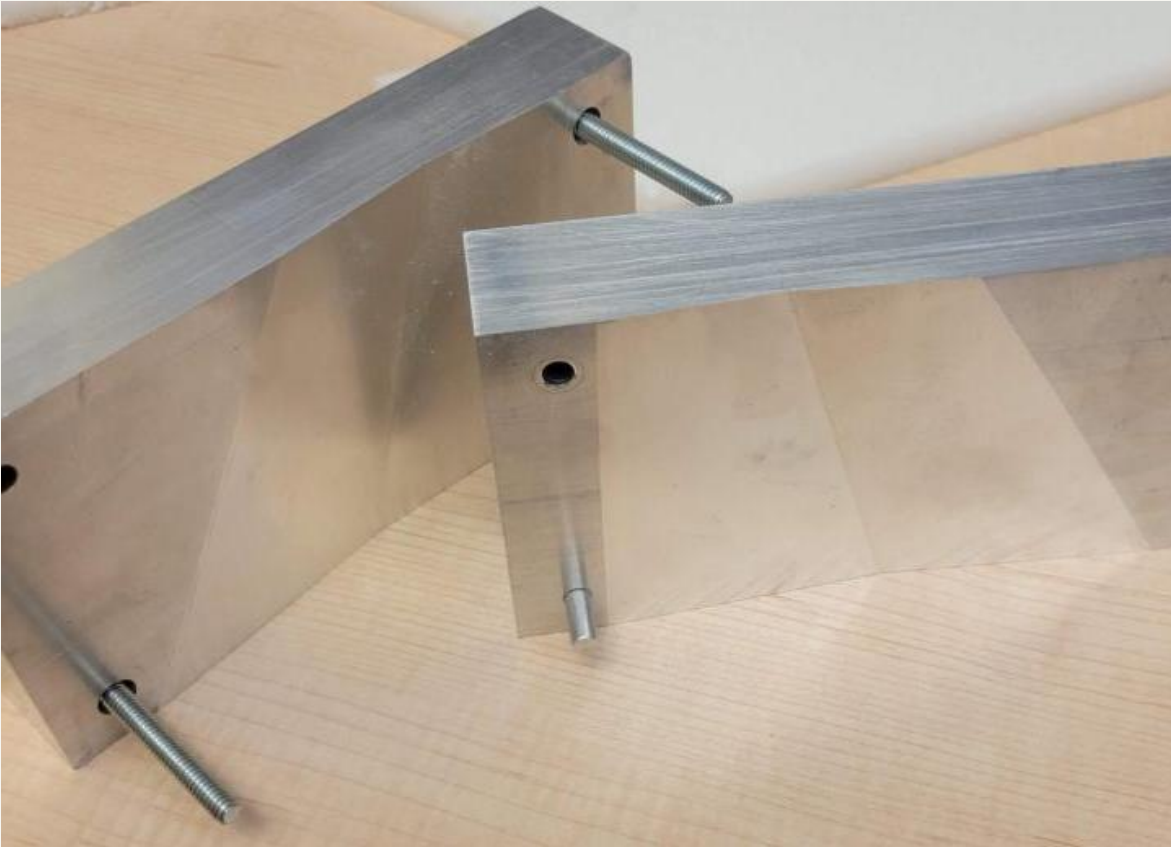
Designed 6 Creo versions for an adaptable gripping fixture. Reduced fasteners from 8 to 1 while improving technician usability and preventing slippage, part damage, and metal-chip FOD. The design passed PDR.

Validated the fixture with hand calculations and Creo static structural FEA.

Preliminary hazard analysis and CONOPS supported formal HAZOP review.

Fin design and thermoforming

04



CNC-machined aluminum tooling

Structured trade study

Evaluated materials, geometries, and manufacturing constraints. Down-selected feasible concepts using analytical feasibility, machinability, and engineering first principles.

Thermoforming process

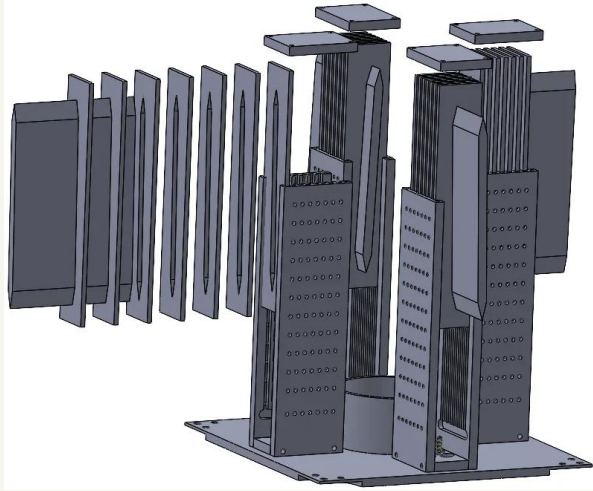
Co-developed and evaluated a process for double-wedge PMI foam fin cores using CNC-machined aluminum tooling and 3D metrology.

0.28 mm

average global geometric error

Fin alignment jig

05



CAD



Machining



Assembly

Roll-coupling problem

1.6° fin cant requirement

Modular four-tower
fixture

CAD, CAM, and machining
fixture

<0.1°

Independent metrology verified all four fins within <0.1° of the 1.6° target.

Designed, CAM programmed, and machined by Nathaniel Murcin

Emergency electrical passthrough

06

During launch integration, the sole flight unit failed.

Designed and fabricated a flight-critical NPT electrical passthrough.

14 hours

design and fabrication turnaround

2× MEOP

hydrostatic test passed

The hardware enabled successful flight.

Composite shear modulus testing

07

TEST OBJECTIVE

Determine the in-plane shear modulus of composite sandwich panels.

EXPERIMENT

Designed and executed a plate-twist testing apparatus.

MODEL VALIDATION

5%

acceptance criterion

The experiment verified the ANSYS model within the acceptance criterion for future fin development.

Manufacturing engineering impact

08

PURSUIT AEROSPACE

Engineering Intern / Aug. 2025 - Dec. 2025

15 quick-change dot-peen fixtures **6 s to 1 s** average load/unload time

40 s to 2 s inspection time using Go/No-Go gauges **55%** less time between completed parts
Broader one-piece-flow changes

3 days less lead time **\$1,000 / part** avoided outsourced cleaning and shipping

Developed an in-house supercleaning process qualified for production use by Rolls-Royce and Pursuit Quality.

Research and publications

09

BEN T. ZINN COMBUSTION LABORATORY

Undergraduate Researcher

Jan. 2026 - Present

Supported reacting jets-in-crossflow experiments by assembling regulated multi-gas delivery hardware and executing established MATLAB/PIV post-processing workflows.

AIAA REGIONAL STUDENT CONFERENCE PROCEEDINGS

2024 **Shear Modulus Testing in Composite Sandwich Panels**

AIAA 2024-85690 / Co-author & presenter

2026 **Viability of Thermoformed Closed-Cell Foam Cores for Carbon Fiber Sandwich Rocket Fins**

AIAA 2026-112793 / Co-author & presenter

Engineering capabilities

10

ANALYSIS

Structural FEA
P&ID design
Cryogenic sizing and safety
Structural and bolted-joint analysis
Flutter analysis

HARDWARE & MANUFACTURING

GD&T and DFM
Composite layups and 3D printing
CNC mill, lathe, and waterjet
Instron testing

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nmurcin.github.io

Appendix: mock launch rail

11



Functional rail for the 2024-2025 GTXR mock test



Two-week delivery

Completed bending and buckling calculations, modeled components in CAD, and manufactured the wood, steel, and aluminum parts.

Result

A functional assembled launch rail for the mock test.